

SANTHOSH REDDY KATASANI VENKATA

Portfolio | LinkedIn | skata9@uic.edu | GitHub | +1 872 899 1118 | Chicago, Illinois

EDUCATION

University of Illinois Chicago

May 2026

Master of Science in Computer Science

Courses: *Big Data Mining, Visual Data Science, Neural Networks, Advanced Machine Learning, ML on Graphs, Deep Learning for Computer Vision, NLP, Pattern Recognition, Computer Algorithms*

SRM Institute of Science and Technology

May 2024

Bachelor of Technology in Computer Science

EXPERIENCE

Research Assistant, UI Health

Jun 2025 – Present

Developed a data-driven recommendation system by implementing ingestion, preprocessing, and vector-based retrieval pipelines over curated healthcare datasets. Designed workflows to clean, transform, and structure nutritional data for efficient querying and downstream analytics. Integrated RESTful APIs using FastAPI to support real-time retrieval and recommendation generation based on user-defined health parameters.

Software Development Intern, Soft Logic Systems

2023 June - 2023 December

Built backend modules for processing application data using Python and MySQL, supporting structured storage and retrieval of transactional records. Implemented data validation and preprocessing workflows, executed SQL-based queries for reporting and debugging, and tested REST APIs using Postman to ensure reliable data exchange across system components.

PROJECTS

1. Cancer Nutrition Analytics Dashboard

Dec 2025

Engineered scalable data ingestion and transformation workflows using public health datasets (NHANES, BRFSS), performing cleaning, normalization, and feature aggregation across demographic and regional attributes. Built structured, query-optimized data models to support cohort-based analysis and derived metric computation, enabling exploratory analytics, performance benchmarking, and downstream reporting workflows.

Technologies: *Python, Pandas, SQL, D3.js*

2. Learning Based Airline Delay Estimation

Oct 2024

Designed distributed data processing pipelines for large-scale flight and weather datasets, implementing structured preprocessing, feature engineering, and schema validation using PySpark. Optimized batch transformations and partitioned data handling to support scalable analytics and performance trend reporting.

Technologies: *Python, PySpark, Pandas, SQL, Tableau*

3. E-Commerce Analytics & Recommendation Platform

Jan 2025

Engineered an end-to-end ELT pipeline ingesting and transforming 100K+ transactional records from the Olist dataset across 9 relational tables into a cloud data warehouse (BigQuery). Designed dbt transformation models to build analytics-ready layers for customer segmentation and order metrics, and orchestrated scheduled workflows using Apache Airflow with automated refresh, logging, and retry mechanisms. Implemented a collaborative filtering recommendation module to support personalized product insights.

Technologies: *Python, SQL, dbt, Airflow, BigQuery*

4. Smart Health Diagnosis Platform

Aug 2024

Developed backend data pipelines to preprocess and map structured symptom datasets into relational schemas for persistent storage and retrieval. Implemented data validation, transformation logic, and normalized MySQL table design to support efficient querying and downstream predictive workflows.

Technologies: *Python, MySQL, SQL, Flask*

TECHNICAL PROFICIENCIES

Languages:	Python, SQL, Java
Data Processing:	Pandas, NumPy, PySpark
Databases:	MySQL, PostgreSQL
Data Analytics:	Exploratory Data Analysis, Data Cleaning, Feature Engineering
Visualization:	Tableau, D3.js
Backend:	FastAPI, Flask
Tools:	Git, Linux